TRINAY CLINICAL CATARACT REPORT

Anterior Segment & Crystalline Lens Triage Examination

ID: CAT-87889

Date: 07 Oct 2026

PATIENT DEMOGRAPHICS & CLINICAL METADATA

Patient Name:	**Walk-in Patient**	Age / Gender:	**55 yrs / Other**	Phone:	**N/A**
Patient ID:	**PX-2879**	Examined Eye:	**Right Eye**	Clinical Operator:	**avi**

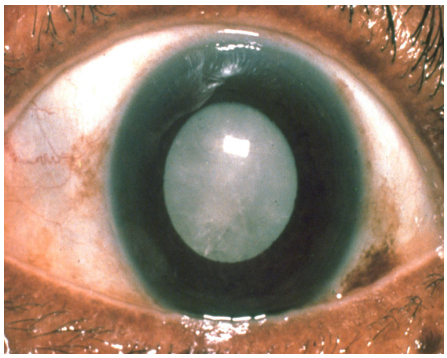
AI SCREENING RESULT: CATARACT DETECTED

Model Probability: 100.0%

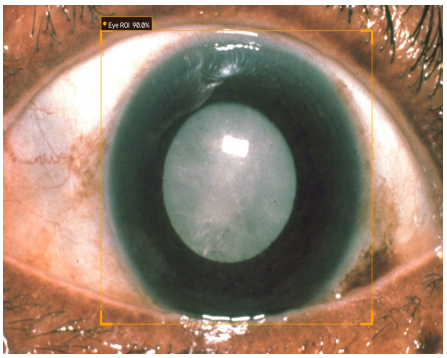
Anterior segment screening indicates lens opacification. Dilated slit-lamp examination advised.

Confidence: 100.0%

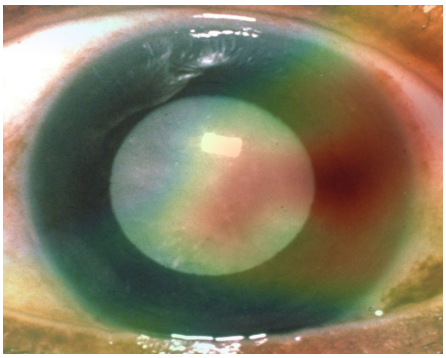
VISUAL AI EVIDENCE



1. Original Eye Image



2. Eye Localization



3. AI Evidence Map — Grad-CAM++

Highlighted regions represent image areas that contributed to the AI classification (interpretability visualization, not definitive lesion proof).

AI CATARACT SCREENING RESULT & CLINICAL ASSESSMENT BREAKDOWN

1. Optical Quality: Gradable (Pass)
Sharp anterior focus, suitable for diagnostic staging

2. Eye Localization: Accurately Centered
Localization confidence: 90.0%

3. AI Evidence Map: Grad-CAM++ Generated
Class-specific activation map overlaid on original eye image

4. Cataract Screening Prediction: Cataract Detected
Lens opacification pattern identified by classifier

5. Model Probability & Confidence: 100.0% Probability
Classification confidence: 100.0% (EfficientNet-V2-S)

6. Triage Recommendation: Referred for Slit-Lamp
Comprehensive ophthalmic slit-lamp biomicroscopy recommended

RECOMMENDED CLINICAL ACTION & REFERRAL TIMELINE

Urgent / Priority referral: Dilated slit-lamp biomicroscopy, visual acuity measurement (Snellen), and intraocular pressure check recommended within 2 to 4 weeks to determine surgical candidacy and prevent visual axis degradation.

DOCTOR'S CLINICAL TRIAGE SUMMARY (FOR ATTENDING CLINICIAN)

Anterior segment screening demonstrates clinically significant lens opacification within the pupillary aperture of the crystalline lens. Pupillary margins are intact without visible coloboma or posterior synechiae. Visual axis compromise is suspected based on focal density and light scattering patterns.

Clinical Findings: Nuclear and/or cortical opacification evident with preserved anterior chamber depth. No evidence of corneal edema or iris neovascularization on primary capture. Recommendation: Schedule dilated slit-lamp biomicroscopy, pinhole visual acuity, and B-scan ultrasonography (if view to fundus is impaired) to classify cataract subtype and determine surgical management.

PATIENT COUNSELING & EXPLANATION (SIMPLE & UNDERSTANDABLE)

What this means for you: A clouding (cataract) was observed in the natural lens of your eye. Just like a frosted window makes it hard to see outside, a cataract scatters incoming light and causes blurry, dimmed, or hazy vision.

Good news: Cataracts are extremely common and can be safely treated. An eye doctor (ophthalmologist) will examine your eyes in detail, test your vision clarity, and discuss treatment options such as updated eyeglasses or a gentle, highly effective outpatient procedure.

CLINICAL PRECAUTIONS & PATIENT GUIDANCE

- Avoid unassisted night driving, especially on roads with bright oncoming headlights that cause glare and halo effects.
- Wear certified UV400 sunglasses during daytime hours to protect the crystalline lens from solar ultraviolet exposure.
- Ensure direct, well-positioned reading lamps and high-contrast lighting when reading or performing detailed tasks.
- Do not rub your eyes and avoid self-prescribing over-the-counter eye drops without formal consultation.
- Schedule an in-person dilated slit-lamp evaluation with an ophthalmologist within the recommended referral timeframe.

CLINICAL VERIFICATION & SIGN-OFF

Triage Screening Nurse:	Screening Platform:	Attending Reviewer Status:
avi	**TRINAY Anterior Vision System**	**Pending Ophthalmologist Review**

REGULATORY & MEDICAL DISCLAIMER

TRINAY Cataract Screening is an AI-assisted diagnostic support tool intended for primary health triage. Definitive clinical diagnosis, cataract staging, and surgical eligibility must be confirmed by an ophthalmologist via in-person slit-lamp biomicroscopy.